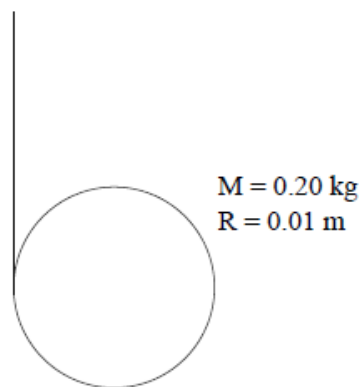


PROBLEMS-Torque

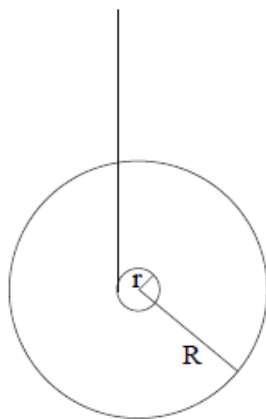
Shown in the figure below is a simple yoyo. The string is wrapped around the outside of the yoyo and the yoyo accelerates downward.



- (a) What is the acceleration of the yoyo?
- (b) What is the tension in the string?

PROBLEMS-Torque

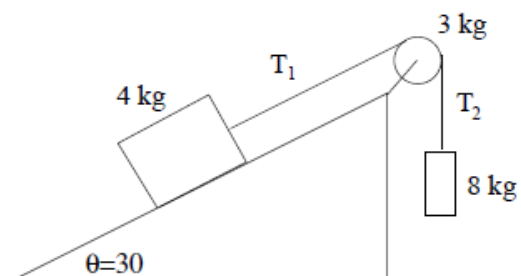
Shown in the figure below is a simple Yoyo. The body of the Yoyo has a radius $R = 0.04\text{ m}$, and the shaft that the string wraps around has radius $r = 0.002\text{ m}$. Approximate the Yoyo as a solid cylinder of radius R and mass $m = 0.100\text{ kg}$.



- (a) (10 pts) What is the downward acceleration of the Yoyo?
- (b) (10 pts) What is the tension in the string?

PROBLEMS-Torque

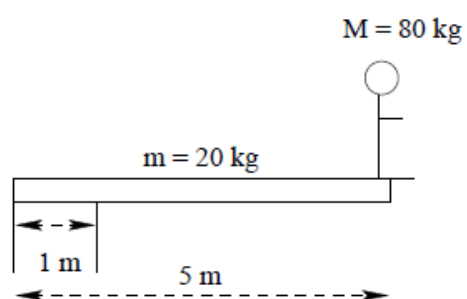
Shown in the figure below is a system of masses and pulleys. NOTE: The pulley has mass and so the tensions on the two sides of the pulley are not the same! Please approximate the pulley as a solid cylinder.



- (a) What is the linear acceleration of the system?
- (b) What is the tension T_1 ?
- (c) What is the tension T_2 ?

PROBLEMS-Torque

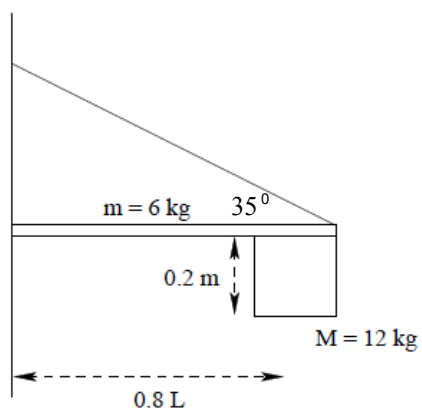
You are standing at the end of a diving board as shown in the figure below.



- (a) What is the force from the “center” support?
- (b) What is the force from the left-most support?

PROBLEMS-Torque

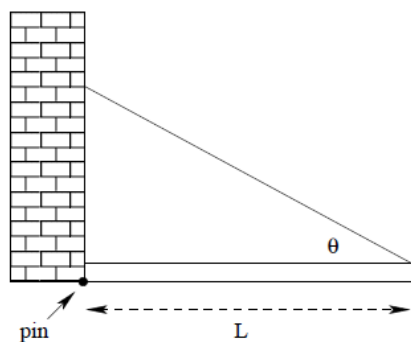
Now that you have your medical degree, you decide to “hang a shingle” (aka a sign) in front of your home and begin your practice.



- (a) What is the tension in the wire?
- (b) What is the normal force from the wall?
- (c) What is the upward force from the wall?

PROBLEMS-Torque

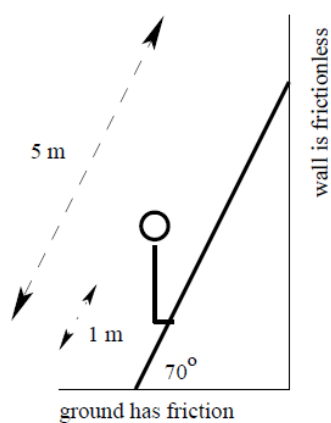
A uniform bar of mass $M = 2.5 \text{ kg}$ and length $L = 2\text{m}$ is supported in static equilibrium in a horizontal position by a pin at its left end attached to the wall, and a string attached at the right end which makes an angle relative to the bar of $\theta = 35^\circ$.



- (a) Find the tension in the string.
- (b) The string breaks and the bar pivots around the pin with no friction. What is its initial angular acceleration, α , just after the string breaks?

PROBLEMS-Torque

Shown in the figure below is a person climbing a ladder. The ladder is 5 meters long and has a mass of 10 kg. The person is standing one meter from the bottom of the ladder and has a mass of 80 kg.



- (a) (6 pts) Find the normal force (N_1) of the ground on the ladder.
- (b) (10 pts) Find the normal force (N_2) of the wall on the ladder.
- (c) (4 pts) Find the frictional force on the bottom of the ladder.